# Liangyu WU | Curriculum Vitae

Tel: (+1) 240-604-1336 | E-mail: liangyu5@stanford.edu

## Education Background

**Stanford University** **California, U.S.**

*Ph.D. Student in Physics* (Advisor: **Dr. Julia Gonski** & **Dr. Dong Su**) Sep. 2024-Present

- Research area: Experimental Particle Physics

**Shanghai Jiao Tong University** **Shanghai, China**

*Bachelor of Science in Physics, School of Physics and Astronomy* Sep. 2020- Jun. 2024

- Overall GPA: 86.1/100 (A-); Physics-related GPA: 90.1/100 (A)
- Coursework: *Quantum Mechanics*, *Electrodynamics*, *Equations of Mathematical Physics*, *Professional Experiment*, *Introduction to Nuclear and Particle Physics*, *Methods of Experimental Nuclear and Particle Physics* (All graded A's)

**University of Maryland, College Park** **Maryland, U.S.**

*Visiting International Student* Aug. 2023-Dec. 2023

- Overall GPA: 3.925/4.0

## Research Experience

**SLAC National Accelerator Laboratory** **California, U.S.**

*Particle Physics and Astrophysics* Jan. 2025-Present

*Research Assistant* (Supervisor: **Dr. Julia Gonski & Dr. Dong Su & Dr. Ariel Schwartzman**)

– **Research on the ATLAS Experiment**

- Foundation model development with ATLAS Run 2 proton-proton collision data.
- Advancing LAr timing for primary vertex $t_0$ reconstruction towards 5D calorimetry in ATLAS.
- GigaBit Cable Receiver (GBCR) ASIC Testing for the HL-LHC ITk upgrade.

– **Research on the Future Colliders**

- On-chip machine learning in future advanced detectors (dual-readout calorimeters, drift chambers).
- R&D for embedded FPGA (eFPGA) technology in advanced subsystems for future colliders.
- Exploration of R&D on dSiPM technology for future detectors.

**SLAC National Accelerator Laboratory** **California, U.S.**

*Particle Physics and Astrophysics* Sep. 2024-Dec. 2024

*Research Assistant* (Supervisor: **Prof. Spencer Gessner**)

– **Research on the FACET-II**

- Enhancements to the X-Band TCAV GUI and associated analysis tools.

**University of Maryland, College Park** **Maryland, U.S.**

*Department of Physics* Aug. 2023-Aug. 2024

*Undergraduate Research Student* (Supervisor: **Prof. Christopher Palmer** & **Prof. Sarah Eno**)

– **Research on the Dual-readout Calorimetry**

- Geant4 simulations of single-particle responses for diverse calorimeters.
- Formula derivation which predicts the dual-readout-corrected energy from scintillator and Cherenkov signals alone.

**Shanghai Jiao Tong University** **Shanghai, China**
*School of Physics and Astronomy* Oct. 2021-Feb. 2024
*Undergraduate Research Assistant* (Supervisor: **Prof. Yue Meng**)

– **R&D for a Novel Radon Detector**

- Designed the Radic detector and determined the radon diffusion coefficient of several membrane materials.

**PandaX Collaboration** **Shanghai & Sichuan, China**
*Undergraduate Research Student* Oct. 2021-Feb. 2023

– **Work in Ultra-low background technique R&D group**

- Measured radon emanation rates across diverse materials for the PandaX and JUNO experiments.
- Conducted ultra-low radioactive surface treatments of materials for TPC assembly in PandaX-4T experiment.

– **Work in PMT group**

- Performed PMT high-voltage testing to ensure standard operating currents and assessed parameters.
- Participated in the installation of veto PMTs and high-voltage cables for TPC-bottom PMTs.

**Tsung-Dao Lee Institute** **Shanghai, China**
*Astronomy and Astrophysics Division* Mar. 2023-Aug. 2023
*Undergraduate Research Student* (Supervisor: **Prof. Masahiro Ogihara**)

– **Research on the stability of unevenly spaced planetary systems**

- Conducted $N$-body simulations to study the planetary system stability.
- Demonstrated that using evenly spaced models can overestimate orbital stability time in natural systems.

## Publications

[1] S. Eno, **L. Wu** *et al.*, On the resolution of dual readout calorimeters, **https://arxiv.org/abs/2501.15329**

[2] S.V. Chekanov, S. Eno, S. Magill, C. Palmer, **L. Wu**, Geant4 simulations of sampling and homogeneous hadronic calorimeters with dual readout for future colliders, **NIM A,Volume 1072, 2025, 170200, ISSN 0168-9002.**

[3] **L. Wu**, L. Si *et al.*, Design and Experimental Application of a Radon Diffusion Chamber for Determining Diffusion Coefficients in Membrane Materials, **JINST, 20, no.03, P03031 (2025)**

[4] S. Yang, **L. Wu** *et al.*, The stability of unevenly spaced planetary systems, **ICARUS, Volume 406, December 2023, 115757**

## Skills

**Technical:** C++, Python, CERN ROOT, Geant4, SolidWorks, COMSOL, LaTeX

**Hardware:** Ultra-low radioactive technology, High-voltage technology, PCB debugging, PMT testing

**Language:** English (Fluent), Mandarin (Native)

## Extra-Curricular Experience

**Student Union of School of Physics and Astronomy** **Shanghai, China**
*Co-President of the Student Union* Dec. 2021-Dec. 2022

– **Spearheaded the execution of events and promotions for the Student Union.**

## Selected Honors and Awards

**2023** Best Report First Prize, TDLI Astro-Division Winter Camp (5 out of 96 participants)

**2022** NetClass Education and Research Fund Scholarship: First Prize, Shanghai Jiao Tong University (Top 5%)

**2021 and 2022** Academic Excellent Scholarship, Shanghai Jiao Tong University (Top 10%, twice)

**2021** Chun Geng Education Scholarship, Shanghai Jiao Tong University, School of Physics and Astronomy (Top 5%)